

Lesson 3 — Break the Big Job into Little Steps

Matches LEGO Teacher Portal lesson: "Caterpillar Friends".

Objective (I can...): I can take one big action and break it into little steps before I code it. **CS concept:** Decomposition (splitting a job into smaller parts) · **Builds on:** Lesson 2 (ordered steps).

AI Connection (2 min, read aloud): To spot a cat, a computer looks at little clues — pointy ears, whiskers, a tail — then puts them together. *Ask the group:* "What little clues tell YOU it's a cat and not a dog?"

Materials: LEGO CS & AI kit + device per pair, big sticky notes or a mini picture-plan mat per pair.

Model reuse: Basics model from Lesson 2 (no rebuild) — code plus one tiny brick tweak.

60-minute flow - Warm-up (10): "Program the human robot" — one child is the robot; the partner gives only tiny steps ("step... step... turn") to walk them to a spot. If a step is too big or fuzzy, the robot freezes with arms out. Point: machines need little, exact steps — that's breaking it down. - **Concept + demo (8):** Coach picks "make it wave hello," says the little steps out loud, then turns each into one picture block. (*Try "the silly robot" — leave out a step and let them catch it.*) - **Build/Code (25):** Using the built model, pairs put one picture-step per sticky note first, then place one block per step and run.

Make-it-yours beat: add one brick or a LEGO face and name the bot. *Wiggle break + swap at ~12 min.*

- **Challenge (7):** Add one more little step (a light or a sound) and show where it goes in the line. - **Share + tidy-up (10):** Pairs show their sticky-note line next to the running, now-named bot. Model stays built.

Pair roles: Navigator lays out the sticky-note steps; Driver codes them. Swap at the wiggle break. (*Trio: third child points to each step as it runs.*)

Differentiation: *Support* — a half-filled step line to finish. *Extension / fast-finisher* — break a two-action job into steps. *Reteach* — act the behavior out with bodies, naming each step, then map to blocks.

Exit check: Pair's sticky-note steps match the blocks in their working program. **At-home (optional):** Break "pour a bowl of cereal" into 5 little steps with a grown-up. ~5 min.